Building quantum codes with transversal CCZ

from projective geometry and SAT solvers

Bohan Lu, Duke University

Joint work with Ken Brown

Virginia Tech Applied Algebra Seminar

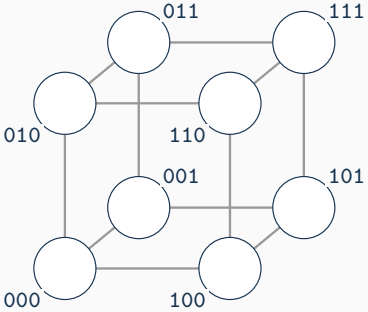
October 22, 2026

arXiv:2610.09341

Outline

1. **CSS codes and transversal gates**
2. The $[\![8, 3, 2]\!]$ code: a worked example of logical CCZ
3. Triorthogonality and the CCZ conditions
4. Projective geometry and the 48-qubit code
5. Column indicators, logical rows, and phase signs
6. Lower bounds and open questions

Binary vectors as Boolean functions

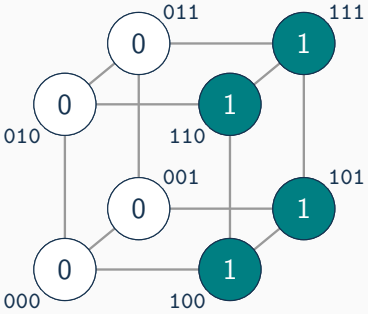


One qubit at each point $u \in \mathbb{F}_2^3$.

Index $n = 8$ qubits by the points $u \in \mathbb{F}_2^3$. A binary vector is a Boolean function, and its weight counts its ones:

$$c\colon \mathbb{F}_2^3 \to \mathbb{F}_2, \qquad |c| = \#\{u : c(u) = 1\}.$$

Binary vectors as Boolean functions



The vector u_1.

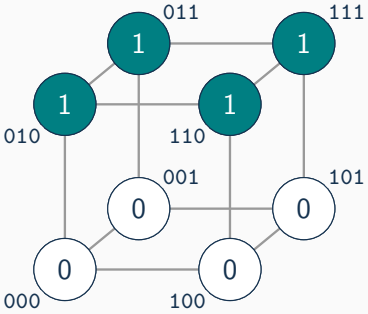
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$$u_1 = 00001111, \qquad u_2 = 00110011.$$

Binary vectors as Boolean functions



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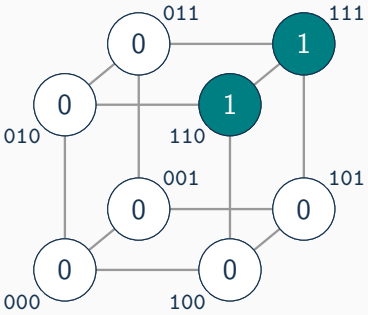
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The overlap $u_1 \wedge u_2$: an edge.

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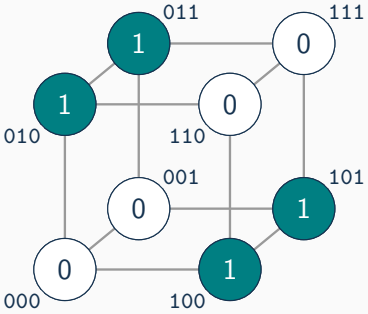
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Overlap $c \wedge c'$ (entrywise product) and dot product $c \cdot c'$:

$$|u_1 \wedge u_2| = 2, \qquad u_1 \cdot u_2 = |u_1 \wedge u_2| \bmod 2 = 0.$$

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The XOR $u_1 \oplus u_2$.

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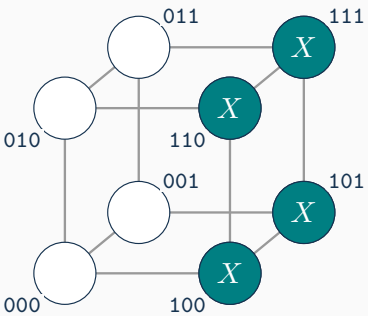
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XOR $c \oplus c'$ (entrywise sum mod 2, not a direct sum):

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Pauli strings

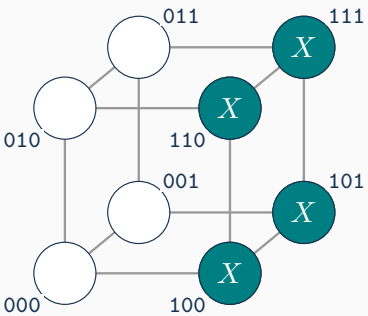


$X(u_1)$: X on the face $u_1 = 1$.

Pauli strings $X(a) = \bigotimes_j X^{a_j}$ and $Z(b) = \bigotimes_j Z^{b_j}$, for $a, b \in \mathbb{F}_2^n$, act on basis states $|c\rangle$:

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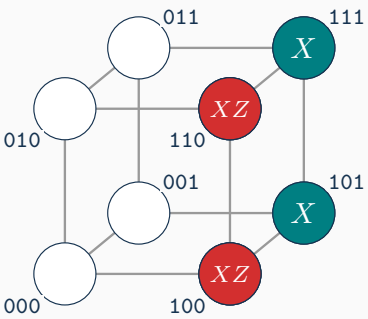
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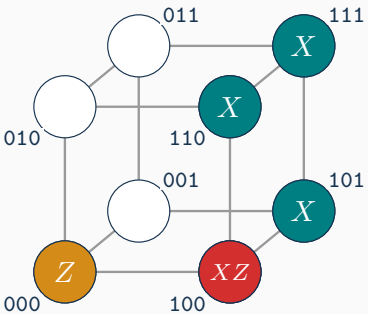
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With $a = u_1$ and δ_u the indicator of the point u:

$$b = \delta_{100} \oplus \delta_{110}: \quad a \cdot b = 0, \text{ commute,}$$

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Stabilizer and CSS codes

Stabilizer code: the joint $+1$ eigenspace of an abelian group of Pauli strings, the *stabilizers*. Flipped eigenvalues form the *syndrome*.

$[\![n, k, d]\!]$: n qubits, dimension 2^k, and d the least weight (number of qubits acted on) of a nontrivial *logical operator*.

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The *CSS code* of $\mathcal{C}_2 \subseteq \mathcal{C}_1 \subseteq \mathbb{F}_2^n$ [CS96, Ste96], with s stabilizer rows S_l and k logical rows K_i, has X-type and Z-type stabilizers

$$X(a),\ a \in \mathcal{C}_2 = \operatorname{rowspan} S, \qquad Z(b),\ b \in \mathcal{C}_1^{\perp}, \qquad \mathcal{C}_1 = \operatorname{rowspan} G,\quad G = \left[\begin{smallmatrix} K \\ S \end{smallmatrix}\right].$$

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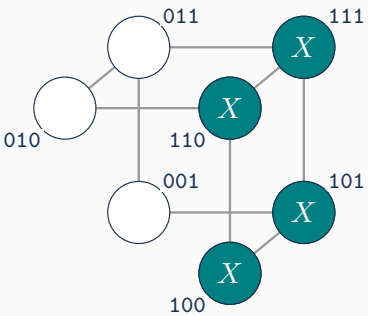
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The logical basis is a uniform superposition over the coset $K^\top x \oplus \mathcal{C}_2$:

$$|x\rangle_L = 2^{-s/2} \textstyle\sum_{y \in \mathbb{F}_2^s} |K^\top x \oplus S^\top y\rangle, \qquad x \in \mathbb{F}_2^k.$$

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The Steane code

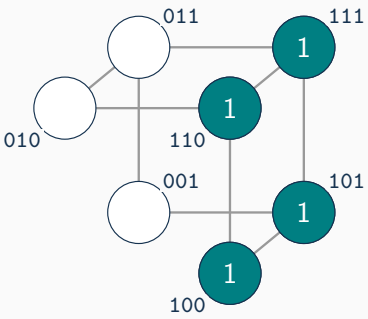


$X(u_1)$, an X-type stabilizer.

Seven qubits at the points $u \neq 000$ [Ste96]. The stabilizer rows are the coordinate functions, and $\mathbf{1}$ is the all-ones vector:

$$S = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}, \qquad K = \mathbf{1}, \qquad \mathcal{C}_1^{\perp} = \mathcal{C}_2.$$

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u_1, a term of $|0\rangle_L$.

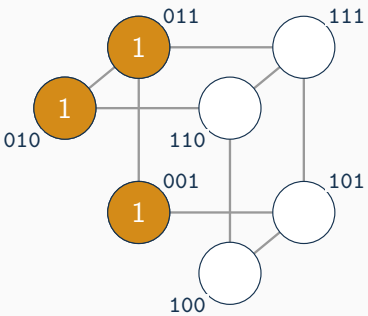
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With $K^{\top} x = x\mathbf{1}$, the logical basis is

$$|x\rangle_L = 2^{-3/2} \sum_{y \in \mathbb{F}_2^3} |x\mathbf{1} \oplus S^{\top} y\rangle, \qquad x \in \mathbb{F}_2.$$

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$\mathbf{1} \oplus u_1$, a term of $|1\rangle_L$.

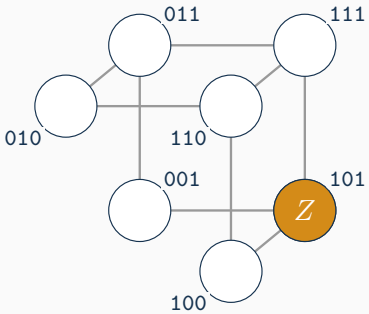
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Syndromes and distance

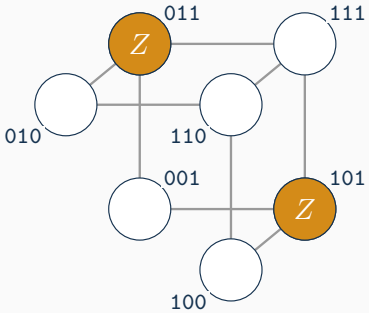


$Z(\delta_{101})$: syndrome 101.

$Z(e)$ anticommutes with $X(S_l)$ exactly when $S_l \cdot e = 1$, so its syndrome is Se. In the Steane code, column u of S is the point u:

$$S\delta_u = u.$$

Syndromes and distance



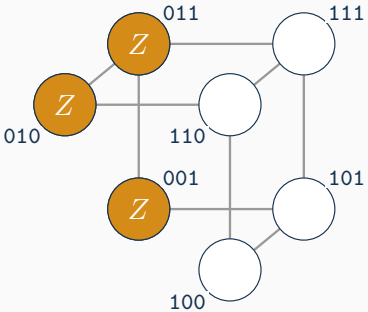
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Syndromes and distance



Z on $001, 010, 011$: syndrome 000.

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Logical operators are $X(v)$, $v \in \mathcal{C}_1 \setminus \mathcal{C}_2$, and $Z(v)$, $v \in \mathcal{C}_2^{\perp} \setminus \mathcal{C}_1^{\perp}$:

$$d_X = \min_{v \in \mathcal{C}_1 \setminus \mathcal{C}_2} |v|, \qquad d_Z = \min_{v \in \mathcal{C}_2^{\perp} \setminus \mathcal{C}_1^{\perp}} |v|.$$

The Steane code has $d_X = d_Z = 3$, so $d = \min(d_X, d_Z) = 3$: a $[\![7, 1, 3]\!]$ code.

Clifford and non-Clifford gates

On basis states, with $x, x_i \in \mathbb{F}_2$:

$$H|x\rangle = \tfrac{1}{\sqrt{2}}\big(|0\rangle + (-1)^x|1\rangle\big), \quad S|x\rangle = i^x|x\rangle, \quad \mathrm{CNOT}|x_1x_2\rangle = |x_1,\, x_1 \oplus x_2\rangle,$$
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S, T and CCZ are *diagonal*, and $|+\rangle = H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$.

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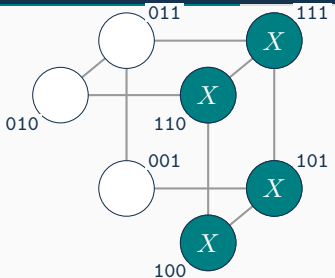
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Clifford gates are not universal. Clifford $+\,T$ and Clifford + CCZ are. A quantum computer needs both kinds as logical gates.

Transversal gates and distillation

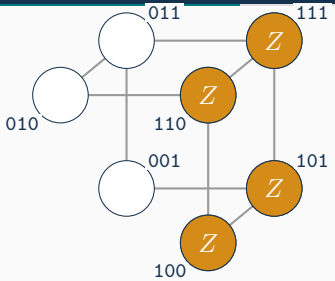


$X(u_1)$.

A logical gate $\bar{U}$ acts on $|x\rangle_L$ as U acts on $|x\rangle$. It is *transversal* if $\bar{U} = U_1 \otimes \cdots \otimes U_n$ on the code space.

[EK09] B. Eastin and E. Knill, Restrictions on transversal encoded quantum gate sets, Phys. Rev. Lett. 102, 110502 (2009), arXiv:0811.4262.
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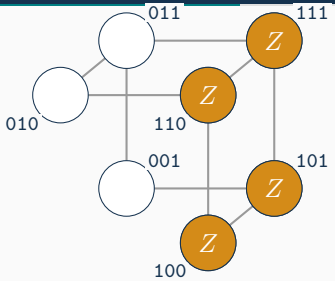
After $H^{\otimes 7}$: $Z(u_1)$.

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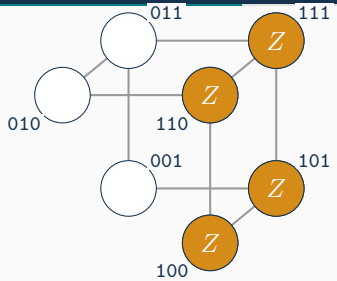
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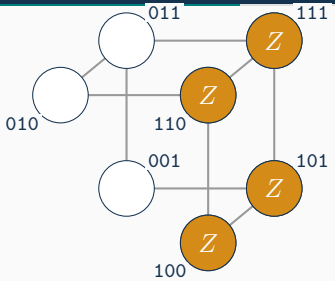
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Magic-state distillation [BK05]: Clifford operations turn many noisy $T|+\rangle$ or $\mathrm{CCZ}|{+}{+}{+}\rangle$ into fewer, less noisy copies, using a code with transversal T or CCZ.

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Transversal gates and distillation



After $H^{\otimes 7}$: $Z(u_1)$.

A logical gate $\bar{U}$ acts on $|x\rangle_L$ as U acts on $|x\rangle$. It is *transversal* if $\bar{U} = U_1 \otimes \cdots \otimes U_n$ on the code space.

Steane: $H^{\otimes 7}$ maps $X(a)$ to $Z(a)$, and $\mathcal{C}_1^\perp = \mathcal{C}_2$, so it is logical H.

H, S, CNOT are transversal on Steane. [EK09]: no code that detects every single-qubit error has a universal transversal gate set, so Steane has no transversal T.

Magic-state distillation [BK05]: Clifford operations turn many noisy $T|+\rangle$ or $\mathrm{CCZ}|{+}{+}{+}\rangle$ into fewer, less noisy copies, using a code with transversal T or CCZ.

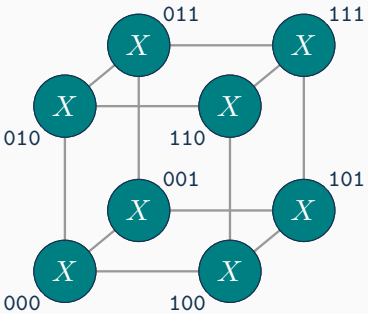
Which CSS codes with $k = 3$ have $U_j \in \{T, T^\dagger\}$ with $U_1 \otimes \cdots \otimes U_n$ logical CCZ?

[EK09] B. Eastin and E. Knill, Restrictions on transversal encoded quantum gate sets, Phys. Rev. Lett. 102, 110502 (2009), arXiv:0811.4262.
[BK05] S. Bravyi and A. Kitaev, Universal quantum computation with ideal Clifford gates and noisy ancillas, Phys. Rev. A 71, 022316 (2005), arXiv:quant-ph/0403025.

Outline

1. CSS codes and transversal gates
2. **The $[\![8, 3, 2]\!]$ code: a worked example of logical CCZ**
3. Triorthogonality and the CCZ conditions
4. Projective geometry and the 48-qubit code
5. Column indicators, logical rows, and phase signs
6. Lower bounds and open questions

The $[\![8, 3, 2]\!]$ code



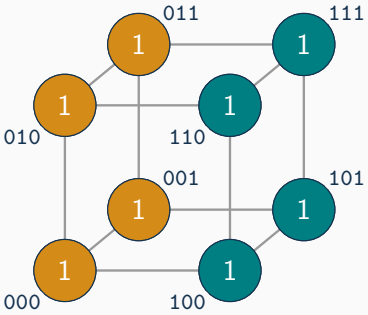
$X(\mathbf{1})$, the X-type stabilizer generator.

Add the origin back and swap the roles of S and K:

$$S = \mathbf{1}, \qquad K = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}, \qquad \mathcal{C}_1 = \mathrm{RM}(1, 3).$$

$\mathrm{RM}(r, m)$: Reed–Muller code, polynomials of degree $\leq r$ in m variables. $\mathrm{RM}(1, 3) = \mathrm{span}\{\mathbf{1}, u_1, u_2, u_3\}$ is an $[8, 4, 4]$ code.

The $[\![8, 3, 2]\!]$ code



$x = 100$. Teal: u_1. Gold: $\mathbf{1} \oplus u_1$.

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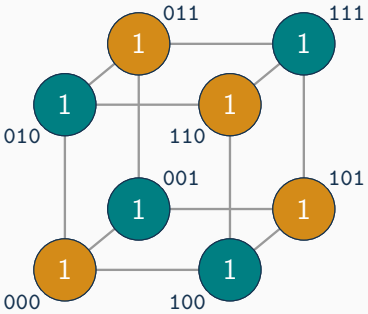
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With $K^\top x = x \cdot u = x_1 u_1 \oplus x_2 u_2 \oplus x_3 u_3$:

$$|x\rangle_L = \big(|x \cdot u\rangle + |\mathbf{1} \oplus x \cdot u\rangle\big)/\sqrt{2}.$$

The $[\![8, 3, 2]\!]$ code



$x = 111$. Teal: $x \cdot u$. Gold: $\mathbf{1} \oplus x \cdot u$.

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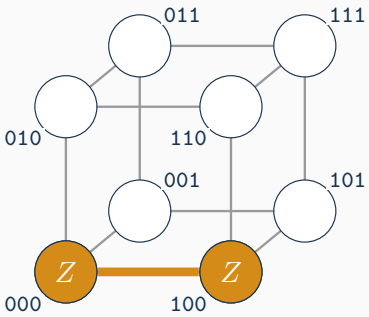
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The $[\![8, 3, 2]\!]$ code



Z on the edge 000–100.

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Z on the edge commutes with $X(\mathbf{1})$:

$$Z(\delta_{000} \oplus \delta_{100})|x\rangle_L = (-1)^{x_1}|x\rangle_L, \qquad d_Z = 2, \quad d_X = 4.$$

Signed weight and the gate condition

A $T/T^\dagger$ *pattern* puts T^{Γ_j} on qubit j, with $\Gamma \in \{\pm 1\}^n$ and the *signed weight* sw_Γ:

$$\bigotimes_j T^{\Gamma_j}|c\rangle = e^{i\pi\,\mathrm{sw}_\Gamma(c)/4}|c\rangle, \qquad \mathrm{sw}_\Gamma(c) = \sum_j \Gamma_j c_j.$$

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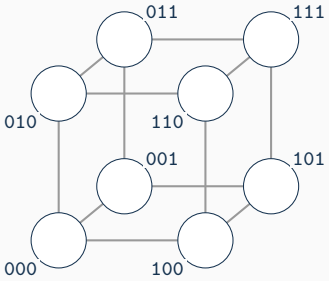
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Logical CCZ: every codeword of the coset $K^\top x \oplus \mathcal{C}_2$ gets the phase $(-1)^{x_1x_2x_3}$:

$$\mathrm{sw}_\Gamma(K^\top x \oplus S^\top y) \equiv 4x_1x_2x_3 \pmod 8 \qquad \text{for all } x \in \mathbb{F}_2^3,\ y \in \mathbb{F}_2^s.$$

The gate condition is a congruence on integer signed weights.

A $T/T^\dagger$ pattern for logical CCZ



XOR of bits as an integer:

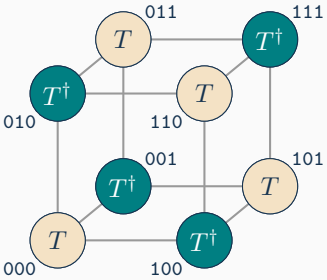
$$a_1 \oplus a_2 \oplus a_3 = \sum_i a_i - 2\sum_{i<j} a_i a_j + 4a_1a_2a_3.$$

Applied at each vertex to $c = K^\top x$, weighted by Γ_u:

$$\begin{aligned}\mathrm{sw}_\Gamma(c) = \sum_i x_i\,\mathrm{sw}_\Gamma(K_i) - 2\sum_{i<j} x_i x_j\,\mathrm{sw}_\Gamma(K_i \wedge K_j) \\ + 4x_1x_2x_3\ \mathrm{sw}_\Gamma(K_1 \wedge K_2 \wedge K_3).\end{aligned}$$

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$T^\dagger$ where $\gamma = u_1 \oplus u_2 \oplus u_3$ is 1 [CH17].

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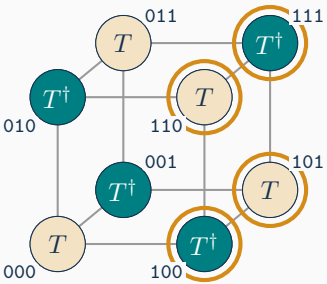
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$K_1 = u_1$: the face $u_1 = 1$.

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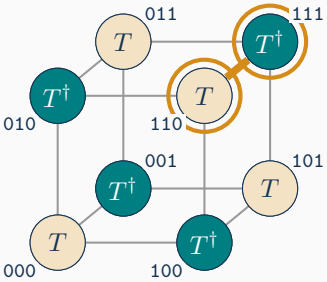
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sw_Γ : face 0,

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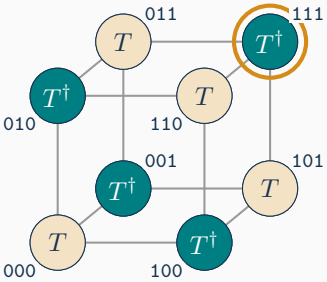
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sw_Γ : face 0, edge 0,

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sw_Γ : face 0, edge 0, vertex 111: -1.

$\mathrm{sw}_\Gamma(c) = -4x_1x_2x_3 \equiv 4x_1x_2x_3 \pmod 8$: logical CCZ.

The gate condition becomes conditions on generator rows

The *Bravyi–Haah parity identity* [BH12], for any binary vectors:

$$\mathrm{sw}_\Gamma(a_1 \oplus \cdots \oplus a_t) = \textstyle\sum_{J \neq \emptyset} (-2)^{|J|-1}\, \mathrm{sw}_\Gamma\big(\bigwedge_{j \in J} a_j\big).$$

Comparing coefficients in the gate condition gives, for distinct rows g, g', g'' of G:

$$\begin{aligned}
\mathrm{sw}_\Gamma(g) &\equiv 0 \pmod 8,\\
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The $[\![8, 3, 2]\!]$ code satisfies all three: faces and edges give 0, and the vertex 111 gives -1.

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Triorthogonal codes and CCZ

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A *triorthogonal matrix* [BH12]: $|g \wedge g'| \equiv |g \wedge g' \wedge g''| \equiv 0 \pmod 2$ for distinct rows g, g', g''.

CCZ needs one odd triple, $K_1 \wedge K_2 \wedge K_3$, so we use the generalized conditions of [CH17] on the next slide.

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Conditions for quasi-transversal CCZ

K1	$\lvert K_i\rvert \equiv 0$
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K3	$\lvert K_1 \wedge K_2 \wedge K_3\rvert \equiv 1$
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Rows K_i of K, S_l of S. All mod 2, indices distinct.

The three row conditions reduce mod 2 to the nine Campbell–Howard (CH) conditions [CH17]:

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A $T/T^\dagger$ pattern with no correction needs the mod 8 and mod 4 conditions on Γ.

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Result: a 48-qubit code with a transversal $T/T^\dagger$ CCZ

The construction [LB26]

$\mathcal{Q}_{48} = [\![48, 3, 3]\!]$, with $(d_X, d_Z) = (16, 3)$. A $T/T^\dagger$ pattern with $26\,T$ and $22\,T^\dagger$ implements logical CCZ with no Clifford correction.

- $\mathcal{C}_2$ is fixed by projective geometry: the columns of S are the points of $\mathrm{PG}(5,2) \setminus \mathrm{PG}(3,2)$, the nonzero vectors of $\mathbb{F}_2^6$ outside a 4-dimensional subspace.
- The logical rows K come from a SAT (Boolean satisfiability) search, and the signs from linear algebra over $\mathbb{Z}_4$.
- Lower bound: no code with $k = 3$, $d_Z \geq 3$, and the nine CH conditions has $n \leq 38$. Lengths 39 to 46 are excluded by [WJS26].
- Removing other subspaces gives codes at 13 block lengths from 48 to 496.

[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).
[WJS26] A. Wills, S. P. Jain, and S. Singh, Classification of generalised triorthogonal codes through length 54, arXiv:2609.30860 (2026).

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4. Solve for the signs Γ over $\mathbb{Z}_4$.

Projectivity rules out weight-one and weight-two Z logicals

Let β_j be column j of S. A Z error on qubit j has syndrome β_j, and Z errors on qubits $i \neq j$ have syndrome $\beta_i \oplus \beta_j$.

- $\beta_j \neq 0$ for all j: every single Z error is detected.
- $\beta_i \neq \beta_j$ for all $i \neq j$: every pair of Z errors is detected.

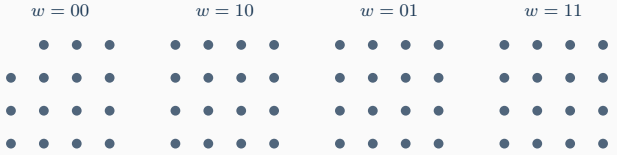
Such S is called *projective*: its columns are distinct points of $\mathrm{PG}(s-1, 2)$, the $2^s - 1$ nonzero vectors of $\mathbb{F}_2^s$. The Steane S is projective: its columns are all 7 points of $\mathrm{PG}(2, 2)$.

$d_Z \geq 3$ follows from the columns of S alone, for every choice of logical rows K.

Complement of a projective subspace

$$\mathcal{P} = \mathrm{PG}(5,2) \setminus \mathrm{PG}(3,2), \qquad |\mathcal{P}| = 63 - 15 = 48.$$

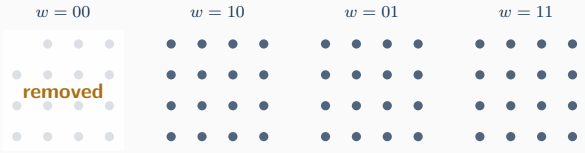
The *support* of S is its set of columns. Write $\beta = (u, w) \in \mathbb{F}_2^4 \times \mathbb{F}_2^2$. The removed $\mathrm{PG}(3,2)$ is the set of 15 nonzero points with $w = 00$. The 48 remaining points form three blocks, one for each $w \neq 00$, with rows $u_1 u_2$ and columns $u_3 u_4$ in Gray order.



Complement of a projective subspace

$$\mathcal{P} = \mathrm{PG}(5,2) \setminus \mathrm{PG}(3,2), \qquad |\mathcal{P}| = 63 - 15 = 48.$$

The *support* of S is its set of columns. Write $\beta = (u, w) \in \mathbb{F}_2^4 \times \mathbb{F}_2^2$. The removed $\mathrm{PG}(3,2)$ is the set of 15 nonzero points with $w = 00$. The 48 remaining points form three blocks, one for each $w \neq 00$, with rows $u_1 u_2$ and columns $u_3 u_4$ in Gray order.

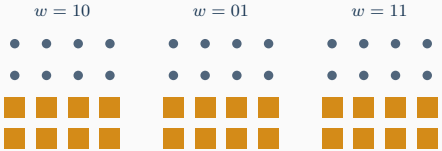


The 6 coordinate functions $u_1, \ldots, u_4, w_1, w_2$ are the rows of S.

Three blocks of 16 points give a projective $[48, 6]$ code $\mathcal{C}_2$.

Hyperplane sections and codeword weights

Each codeword c of $\mathcal{C}_2$ is a linear function $c(u, w) = \lambda \cdot u \oplus \mu \cdot w$ on $\mathcal{P}$, and $|c| = 48 - |\mathcal{P} \cap \{c = 0\}|$.



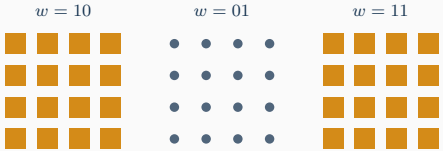
$\lambda \neq 0$, for example $c = u_1$: eight points in each block, weight $3 \cdot 8 = 24$.

Projective *two-weight* codes (two nonzero weights) correspond to point sets that meet every hyperplane in one of two sizes [CK86].

[CK86] A. R. Calderbank and W. M. Kantor, The geometry of two-weight codes, Bull. London Math. Soc. 18, 97 (1986), doi:10.1112/blms/18.2.97.

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$\lambda = 0$, for example $c = w_1$: two full blocks, weight $2 \cdot 16 = 32$.

Projective *two-weight* codes (two nonzero weights) correspond to point sets that meet every hyperplane in one of two sizes [CK86].

Weights 24 and 32 are multiples of 8, so S1–S3 hold.

[CK86] A. R. Calderbank and W. M. Kantor, The geometry of two-weight codes, Bull. London Math. Soc. 18, 97 (1986), doi:10.1112/blms/18.2.97.

More geometric constructions

Choose a subspace complement

$$\mathrm{PG}(p+q-1,2) \setminus \mathrm{PG}(p-1,2), \qquad n = 2^{p+q} - 2^p = 2^p(2^q - 1).$$

The paper constructs codes at 13 block lengths from 48 to 496, with $d_Z = 3$ or 4.

$p \backslash q$	1	2	3	4	5
4	none	$[\![48,3,3]\!]$, 2*	$[\![112,3,3]\!]$, 12	$[\![240,3,3]\!]$, 17	$[\![496,3,3]\!]$, 18
5	none	$[\![96,3,3]\!]$, 11	$[\![224,3,3]\!]$, 8	$[\![480,3,3]\!]$, 14	
6	$[\![64,3,4]\!]$, 5	$[\![192,3,3]\!]$, 14	$[\![448,3,3]\!]$, 13		
7	$[\![128,3,4]\!]$, 3	$[\![384,3,3]\!]$, 18			
8	$[\![256,3,4]\!]$, 3				

Each cell: parameters with $d = d_Z < d_X$, and the number of inequivalent codes found. Counts are lower bounds except $\star$ (complete). "None": excluded by our lower bound. Supports with $p \leq 3$ admit no $T/T^\dagger$ implementation of CCZ.

Outline

1. CSS codes and transversal gates
2. The $[\![8, 3, 2]\!]$ code: a worked example of logical CCZ
3. Triorthogonality and the CCZ conditions
4. Projective geometry and the 48-qubit code
5. **Column indicators, logical rows, and phase signs**
6. Lower bounds and open questions

Column indicators from the generator matrix

Let M have distinct columns in $\mathbb{F}_2^r$. Its *column indicator* [NH22] is the Boolean function on $\mathbb{F}_2^r$ with $\chi_M(v) = 1$ exactly when v is a column of M. We write χ_M as a multilinear polynomial over $\mathbb{F}_2$. Its weight $|\chi_M|$ is the number of columns of M.

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For the $[\![8, 3, 2]\!]$ code, $G = [K; S]$ has the columns $(u, 1)$, $u \in \mathbb{F}_2^3$:

$$G = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

In coordinates $(x, y) \in \mathbb{F}_2^3 \times \mathbb{F}_2$, with x for the rows of K and y for the row of S, the columns are exactly the points with $y = 1$, so $\chi_G(x, y) = y$.

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In coordinates $(x, y) \in \mathbb{F}_2^3 \times \mathbb{F}_2$, with x for the rows of K and y for the row of S, the columns are exactly the points with $y = 1$, so $\chi_G(x, y) = y$.

$\mathcal{Q}_{48}$: the columns of S are the points with $w \neq 00$, so

$$\chi_S(u, w) = 1 + (1 + w_1)(1 + w_2) = w_1 + w_2 + w_1 w_2.$$

The CH conditions are inner products with monomials

For Boolean functions on $\mathbb{F}_2^r$, let $\langle f, f' \rangle$ be the dot product of their value vectors [CH17]. Index the rows of G by coordinates $z = (x, y)$. For a set J of rows, $m_J = \prod_{j \in J} z_j$ and

$$\langle \chi_G, m_J \rangle = \left| \bigwedge_{j \in J} g_j \right| \bmod 2.$$

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Since $\mathrm{RM}(3, r)^{\perp} = \mathrm{RM}(r-4, r)$, and $y_1 \cdots y_s$ (one at the 8 points $(x, 1, \ldots, 1)$) satisfies these conditions [LB26],

$$\chi_G \in y_1 \cdots y_s + \varepsilon \delta_0 + \mathrm{RM}(r-4, r), \qquad r = 3 + s,\ \varepsilon = n \bmod 2.$$

Logical rows in the column picture

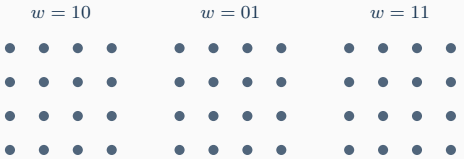
Column $\beta = (u, w) \in \mathcal{P}$ of S has the *logical label* $\kappa(\beta) \in \mathbb{F}_2^3$, the corresponding column of K [LB26]:

$$\chi_G(x, u, w) = \chi_S(u, w) \prod_{i=1}^{3} \big(1 + x_i + \kappa_i(u, w)\big).$$

Split a set J of rows into its logical rows J_K and stabilizer rows J_S. The CH conditions become

$$\bigoplus_{\beta \in \mathcal{P}} m_{J_K}(\kappa(\beta))\, m_{J_S}(\beta) = \begin{cases} 1, & J = \{K_1, K_2, K_3\}, \\ 0, & \text{else.} \end{cases}$$

S1–S3 have $J_K = \emptyset$ and hold because $\mathcal{C}_2$ is 8-divisible. The rest have degree at most 3 in the label bits.



Logical rows in the column picture

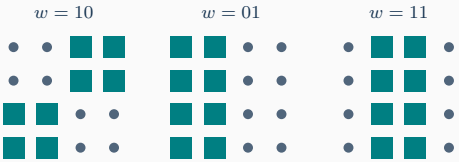
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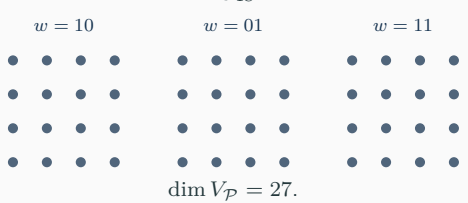
Teal: $\kappa_1 = 1$, with $\kappa_1 = u_1 + u_3$, $1 + u_3$, u_4 on the three blocks.

Finding the logical rows

Fix $\mathcal{P}$. Row K_i is $\kappa_i \colon \mathcal{P} \to \mathbb{F}_2$, bit i of the label map, and $\langle f, f' \rangle_{\mathcal{P}} = \sum_{\beta \in \mathcal{P}} f(\beta) f'(\beta) \bmod 2$ [LB26].

1. K1, KS1, KS3 say $\kappa_i \perp \mathbf{1}, y_l, y_l y_{l'}$ (y_l: row l of S), which span $\mathrm{RM}(2, s)$:

$$\kappa_i \in V_{\mathcal{P}} = \{ f : \langle f, m \rangle_{\mathcal{P}} = 0,\ m \in \mathrm{RM}(2, s) \}.$$



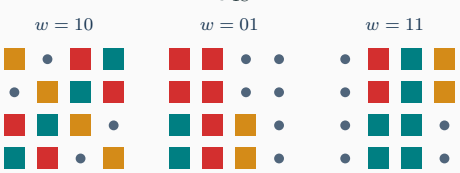
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2. K2, KS2: choose a pair with $\langle \kappa_1 \kappa_2, \mathbf{1} \rangle_{\mathcal{P}} = \langle \kappa_1 \kappa_2, y_l \rangle_{\mathcal{P}} = 0$.



Teal: only K_1 has a 1. Gold: only K_2 has a 1. Red: both. $|K_1 \wedge K_2| = 12 \equiv 0 \pmod 2$ (K2).

Finding the logical rows

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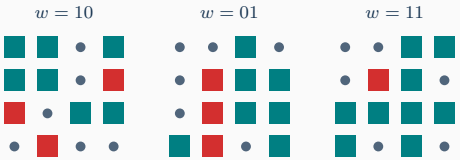
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3. The rest is linear in $\kappa_3 \in V_{\mathcal{P}}$, solved by Gaussian elimination:

$$\langle \kappa_3, \kappa_i \rangle_{\mathcal{P}} = \langle \kappa_3, y_l \kappa_i \rangle_{\mathcal{P}} = 0,\ \langle \kappa_3, \kappa_1 \kappa_2 \rangle_{\mathcal{P}} = 1.$$

$\mathcal{Q}_{48}$



Squares: K_3 has a 1. Red: K_1 and K_2 too.
$|K_1 \wedge K_2 \wedge K_3| = 7 \equiv 1 \pmod 2$ (K3).

Finding the logical rows

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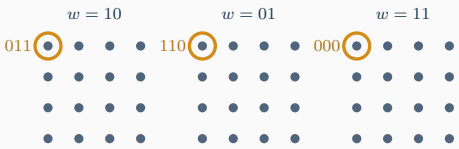
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   If no pair extends, $\mathcal{P}$ is excluded.

$\mathcal{Q}_{48}$

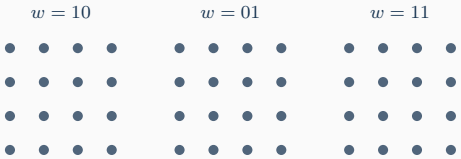


Z on the circled qubits, at $u = 0000$ (the upper left corner of each block). Their columns of S sum to 0, so no syndrome. Their labels sum to $011 \oplus 110 \oplus 000 = 101 \neq 0$, so it is logical: $d_Z = 3$.

Solving for the $T/T^\dagger$ pattern

As on the cube, $\Gamma_j = 1 - 2\gamma_j$ with $\gamma_j = 1$ on the $T^\dagger$ qubits. Then $\mathrm{sw}_\Gamma(c) = |c| - 2|\gamma \wedge c|$, and every codeword $c = K^\top x \oplus S^\top y$ has even weight:

$$|\gamma \wedge c| \equiv \tfrac{1}{2}|c| - 2x_1x_2x_3 \pmod 4.$$



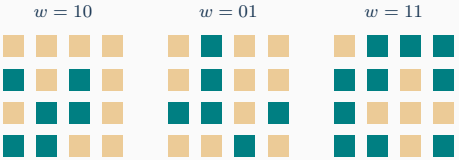
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$$|\gamma \wedge c| \equiv \tfrac{1}{2}|c| - 2x_1x_2x_3 \pmod 4.$$

For $\mathcal{Q}_{48}$ [LB26]:

$$\begin{aligned}\gamma(u,w) &= u_3 + w_1 Q_1(u) + w_2 Q_2(u),\\ Q_1 &= u_1 + u_2 + u_3 + u_1u_3 + u_2u_3 + u_2u_4,\\ Q_2 &= u_3 + u_4 + u_1u_2 + u_1u_4 + u_3u_4.\end{aligned}$$



Teal: $T^\dagger$ $(6, 6, 10$ per block). Gold: T. Total $26\,T + 22\,T^\dagger$.

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4. Projective geometry and the 48-qubit code
5. Column indicators, logical rows, and phase signs
6. **Lower bounds and open questions**

Stabilizer supports are Reed–Muller codewords

For projective S, S1–S3 say $\langle \chi_S, m \rangle = 0$ for every monomial m in $y_1, \ldots, y_s$ of degree 1 to 3.

Following [NH22], let $c = \chi_S + \varepsilon \delta_0$, where ε makes $|c|$ even. Every monomial of degree at least 1 vanishes at 0, so adding δ_0 changes only $\langle c, \mathbf{1} \rangle = |c| \bmod 2$. Hence $c \perp \mathrm{RM}(3, s)$ and

$$c \in \mathrm{RM}(3, s)^{\perp} = \mathrm{RM}(s - 4, s), \qquad n = |c| - c(0).$$

For $\mathcal{Q}_{48}$: $c = \chi_S = w_1 + w_2 + w_1 w_2 \in \mathrm{RM}(2, 6)$.

Our exclusion argument covers every length through 38

1. Reduce the stabilizer rank s (the number of rows of S) to $s \leq 8$ when $n \leq 38$.
2. Exclude $s \leq 5$, then settle ranks 6 and 7 using the possible weights and the geometry of the support.
3. At rank 8, use the classification of low-weight Reed–Muller codewords to reduce to finitely many supports, and enumerate them.
4. Show that no remaining support admits logical rows satisfying K1–K3 and KS1–KS3.

Independent result of our paper

No CSS code with $k = 3$, $d_Z \geq 3$, and the nine CH conditions has $n \leq 38$.

Existence of a valid code at different block lengths

Length	Status	Evidence
$n \leq 38$	Excluded	Our lower bound
$39 \leq n \leq 46$	Excluded	[WJS26]
$n = 47$	Quasi-transversal CCZ	[JVB+26]
$n = 48$	Physical $T/T^\dagger$, no correction	Our explicit code

Wills–Jain–Singh classify generalized triorthogonal codes with $n \leq 54$ and $d_Z \geq 3$, which covers quasi-transversal CCZ with $k = 3$.

The $[\![47, 3, 3]\!]$ code of [JVB+26] has no $T/T^\dagger$ pattern [LB26]. Its stabilizer support is $\mathbb{F}_2^6$ minus an affine 4-flat (a translate of a 4-dimensional subspace), with the origin deleted. $\mathcal{Q}_{48}$ removes a linear 4-dimensional subspace instead. [JVB+26] report a 47-to-1 CCZ distillation protocol from it with a circuit footprint of 9 qubits.

[JVB+26] H. Jacinto et al., Exploring the landscape of compact magic-state distillation factories, arXiv:2606.07734 (2026).

Takeaways and open questions

Takeaways

- Projective geometry fixes S: $d_Z \geq 3$ and weights divisible by 8.
- Column indicators turn the CH conditions into Boolean equations. The signs solve a $\mathbb{Z}_4$ system.
- $\mathcal{Q}_{48} = [\![48, 3, 3]\!]$ with $26\,T + 22\,T^\dagger$. Our bound excludes $n \leq 38$, and [WJS26] excludes 39–46.

Open questions

- Which supports admit logical rows K and signs Γ?
- Higher distance, other gates, more logical qubits.
- Minimum length for $T/T^\dagger$ CCZ with no correction.
- Full circuits, noise, and resource costs.

Paper
arXiv:2610.09341

Code and data
github.com/jerrylvx/gate2code-public

Backup: admissible lengths from Reed–Muller weights

S1–S3 hold if and only if $\chi_S \in \mathrm{RM}(s-4,s) \cup \big(\delta_0 + \mathrm{RM}(s-4,s)\big)$ [LB26]. At rank s the lengths are $n = |c| - c(0)$ for $c \in \mathrm{RM}(s-4,s)$ whose support spans $\mathbb{F}_2^s$.

s	code	weights ≤ 39	candidate n, $15 \leq n \leq 38$
5	$\mathrm{RM}(1,5)$	16, 32 [MS77]	15, 16, 31
6	$\mathrm{RM}(2,6)$	16, 24, 28, 32, 36 [SB70]	15, 16, 23, 24, 27, 28, 31, 32, 35, 36
7	$\mathrm{RM}(3,7)$	16, 24, 28, 32, 36 [KT70, Car21]	the same as $s = 6$

[MS77] F. J. MacWilliams and N. J. A. Sloane, The Theory of Error-Correcting Codes, North-Holland (1977).

[SB70] N. J. A. Sloane and E. R. Berlekamp, Weight enumerator for second-order Reed–Muller codes, IEEE Trans. Inf. Theory 16, 745 (1970).

[KT70] T. Kasami and N. Tokura, On the weight structure of Reed–Muller codes, IEEE Trans. Inf. Theory 16, 752 (1970).

[Car21] C. Carlet, Boolean Functions for Cryptography and Coding Theory, Cambridge University Press (2021), Sec. 4.1.5 (Ax's theorem).

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7	$\mathrm{RM}(3,7)$	16, 24, 28, 32, 36 [KT70, Car21]	the same as $s = 6$

$n = 15, 16$: then $|c| = 16$, the minimum weight, so the support is an affine 4-flat. On a flat of dimension $m \leq 4$, K1, KS1, KS3 force $\deg \kappa_i \leq m-3$, so $\deg(\kappa_1\kappa_2\kappa_3) \leq 3(m-3) \leq m-1$. Then $|K_1 \wedge K_2 \wedge K_3|$ is even, which contradicts K3.

[MS77] F. J. MacWilliams and N. J. A. Sloane, The Theory of Error-Correcting Codes, North-Holland (1977).

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Backup: lowering the stabilizer rank

Project along a nonzero $v \in \mathbb{F}_2^s$ (kernel $\{0, v\}$) and delete one row S_m with $v_m = 1$, after $K_i \mapsto K_i \oplus \delta_i S_m$. The label differences along v are

$$\Delta_v = \{\kappa(\beta \oplus v) \oplus \kappa(\beta) : \ \beta, \ \beta \oplus v \in \mathcal{P}\} \subseteq \mathbb{F}_2^3.$$

If $\Delta_v \subseteq \{\delta\}$, and $\kappa(v) = \delta$ when $v \in \mathcal{P}$, the result keeps the nine CH conditions, $k = 3$ and $d_Z \geq 3$ at rank $s - 1$ [LB26].

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At most $\lfloor \binom{n}{2}/2 \rfloor$ vectors v have $|\Delta_v| \geq 2$, and n lie in $\mathcal{P}$, so a good v exists when

$$2^s - 1 > n + \left\lfloor \tbinom{n}{2}/2 \right\rfloor. \qquad n \leq 38 : \ 38 + 351 = 389 < 2^9 - 1, \text{ so } s \leq 8.$$

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$s \leq 5$: of the 63 supports from $\mathrm{RM}(1, 5)$, 62 are hyperplanes or punctured hyperplanes, which are flats. The last, $\mathbb{F}_2^5 \setminus \{0\}$ with $n = 31$, fails Algorithm 1 on all $49{,}741{,}825$ pairs.

Backup: the finite cases at ranks 6, 7 and 8

s	$\lvert c\rvert$	cases	result
6, 7	16	affine 4-flat	K3 fails
6	24–36	six affine classes (Dickson forms). The degree-one class lies in a hyperplane, so the rank drops to 5	no κ_3
7	24, 28	1 and 2 classes [KT70]	no κ_3
7	32	y_1y_2 (in a hyperplane) and 3 cubic classes [KTA76]	no κ_3
7	36	9,792 functions, 7,038 supports	no κ_3
8	≤ 30	counting: $n + \lfloor \binom{n}{2}/2 \rfloor \leq 247 < 2^8 - 1$	rank drops
8	32, $\deg c \leq 3$	affine 5-flat, rank ≤ 6	excluded
8	32–38, $\deg c = 4$	$c = y_7 f + y_8 g + y_7 y_8 h$ with $\lvert c\rvert = \lvert f\rvert + \lvert g\rvert + \lvert f + g + h\rvert$, $(\lvert f\rvert, \lvert g\rvert) \in \{(8,8),(8,12),(12,12),(8,14)\}$, 2,399 orbit representatives of g, 469,728 supports	no κ_3

Every case ends in a flat or hyperplane reduction, or in Algorithm 1 finding no third logical row.

[KTA76] T. Kasami, N. Tokura, and S. Azumi, On the weight enumeration of weights less than 2.5d of Reed–Muller codes, Inf. Control 30, 380 (1976).

Backup: dictionary with [CLM+26]

[CLM+26]	This talk
$U(b) = \bigotimes_j \mathrm{diag}(1, \omega^{b_j})$, $\omega = e^{2\pi i/N}$, $N = 2^\ell$	$N = 8$, $b \equiv \Gamma$: $T = U(1)$, $T^\dagger = U(-1)$
$b \cdot w$ for $w \in \mathbb{F}_2^n$	$\mathrm{sw}_\Gamma(w)$
$U(b)$ fixes the code iff	rows with a stabilizer row: $\mathrm{sw}_\Gamma(S_l) \equiv 0$ (8),
$b \in H_N = \left(\beta_2 \star \bigcup_{i<\ell} 2^i \beta_1^{(i)}\right)^{\perp_N}$	$\mathrm{sw}_\Gamma(S_l \wedge g) \equiv 0$ (4), $\mathrm{sw}_\Gamma(S_l \wedge g \wedge g') \equiv 0$ (2)
logical action $\omega^{b\cdot(Bv)} = \prod_J C_J\text{-}U(a_J)$,	CCZ $= C_{123}\text{-}U(4)$: $\mathrm{sw}_\Gamma(K_i) \equiv 0$ (8),
$a_J = (-2)^{\lvert J\rvert-1}\, b \cdot (w_{j_1} \star \cdots \star w_{j_h})$	$\mathrm{sw}_\Gamma(K_i \wedge K_{i'}) \equiv 0$ (4),
	$\mathrm{sw}_\Gamma(K_1 \wedge K_2 \wedge K_3) \equiv 1$ (2)

Our signed conditions are their H_8 with $b \in \{\pm 1\}^n$ and the logical action fixed to CCZ. We solve the converse problem: given the action, find S, K and b.